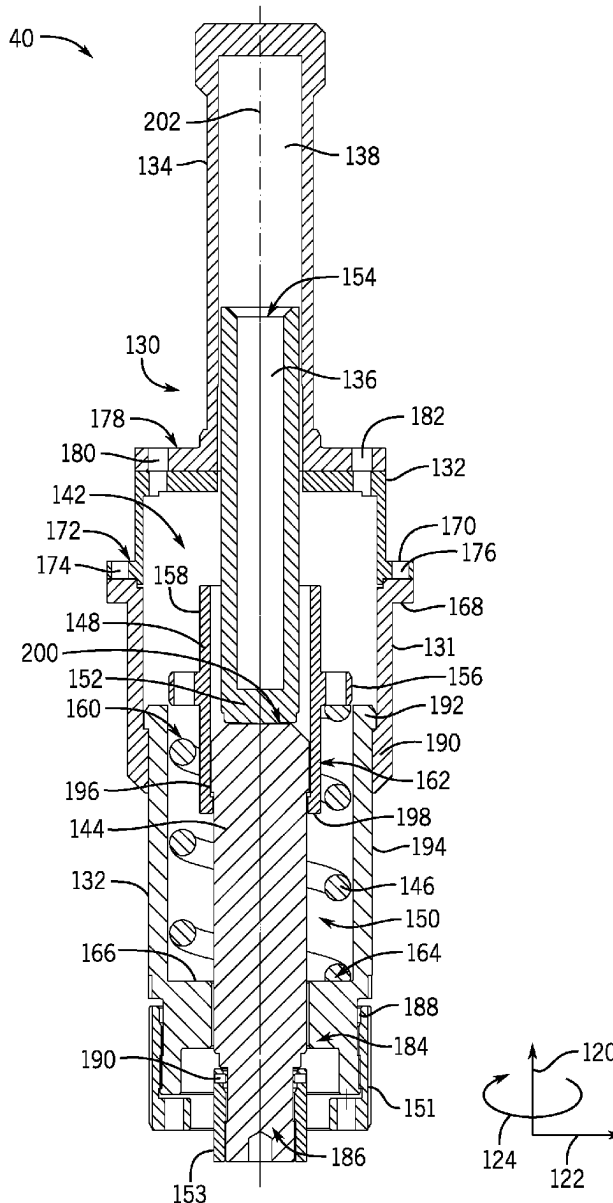


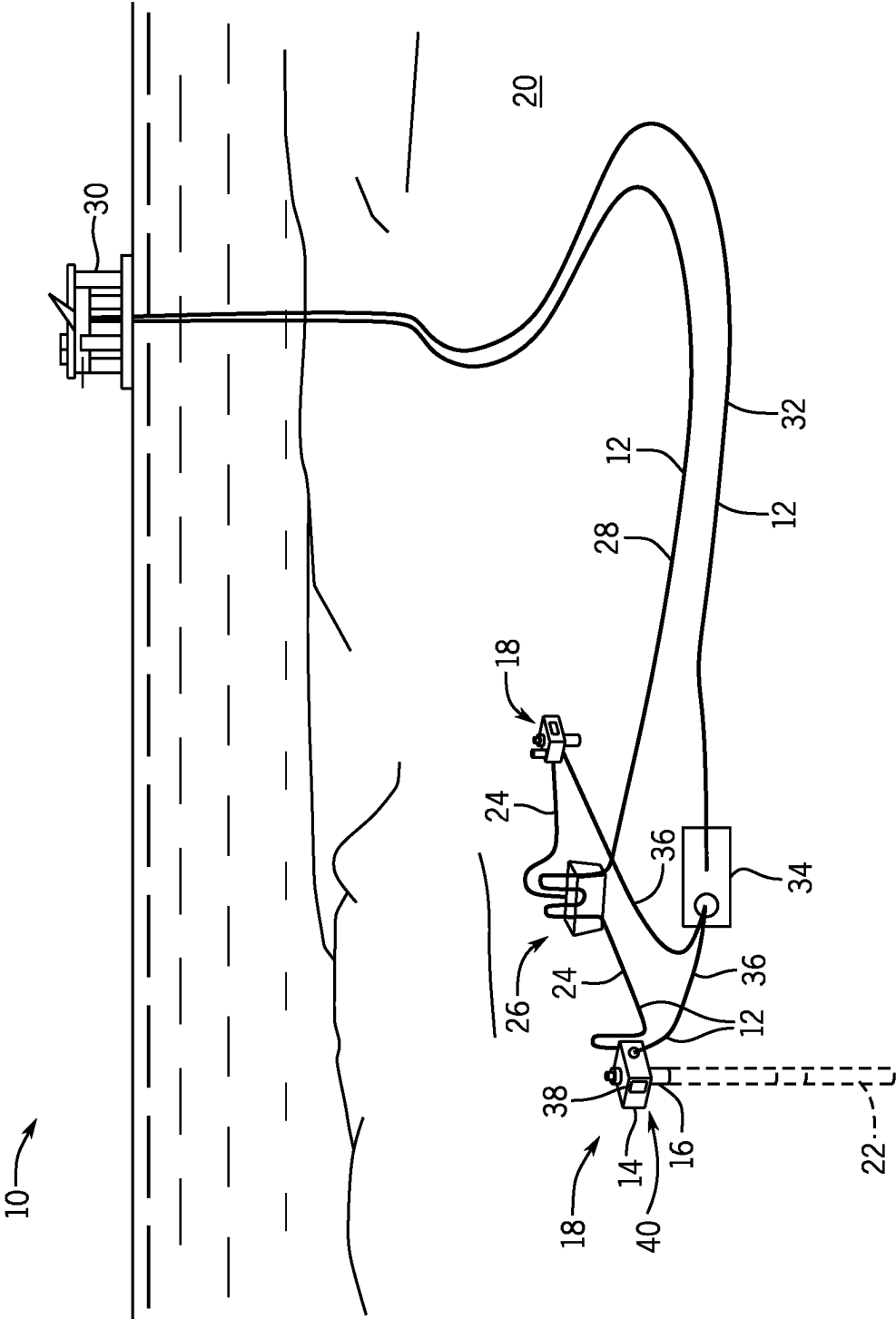


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Fendeler De Lopes**, Macae (BR)(57) **ABSTRACT**

A system includes a landing system configured to land a retrievable module on a subsea structure. The landing system includes a landing post, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.





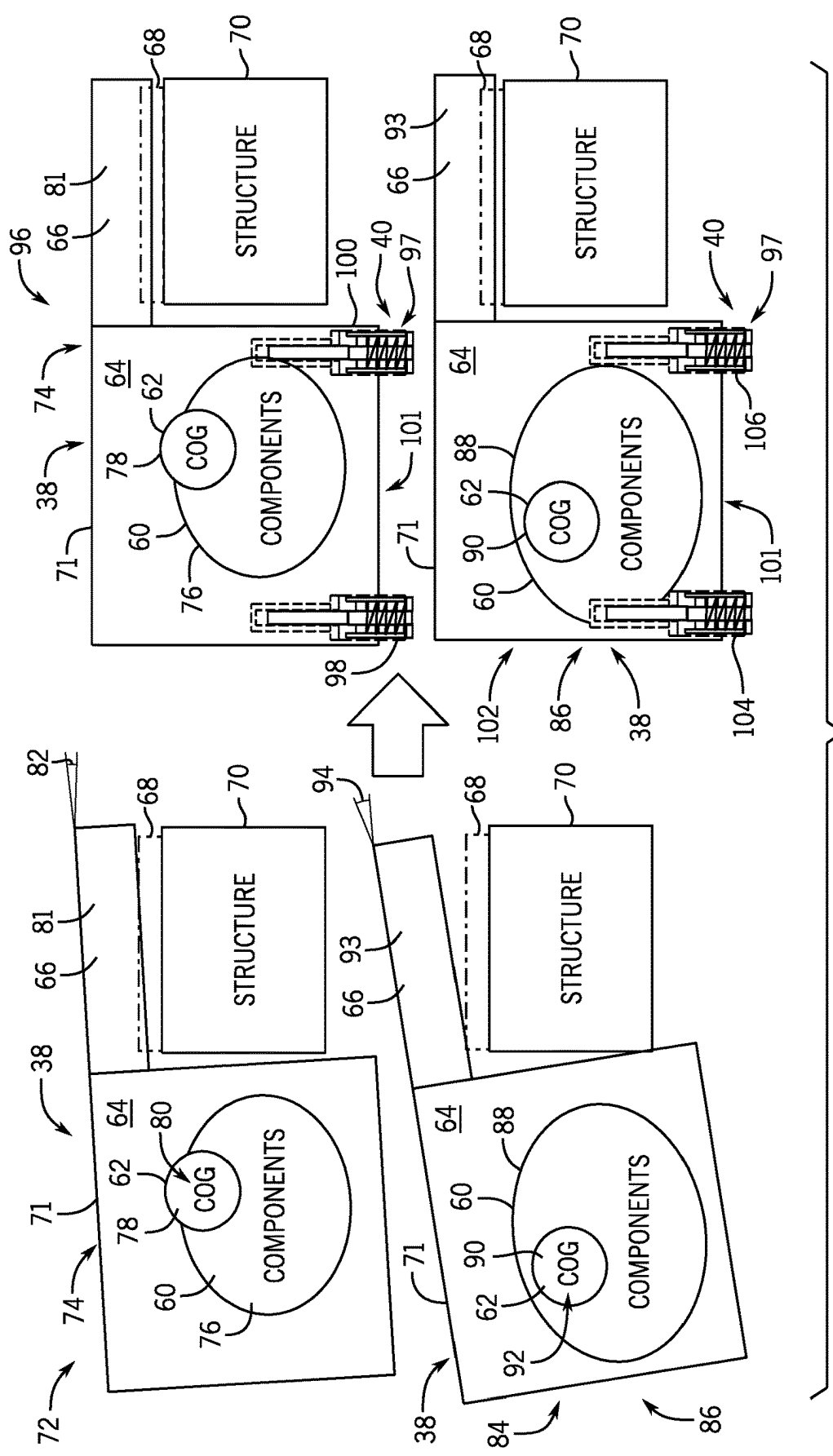


FIG. 2

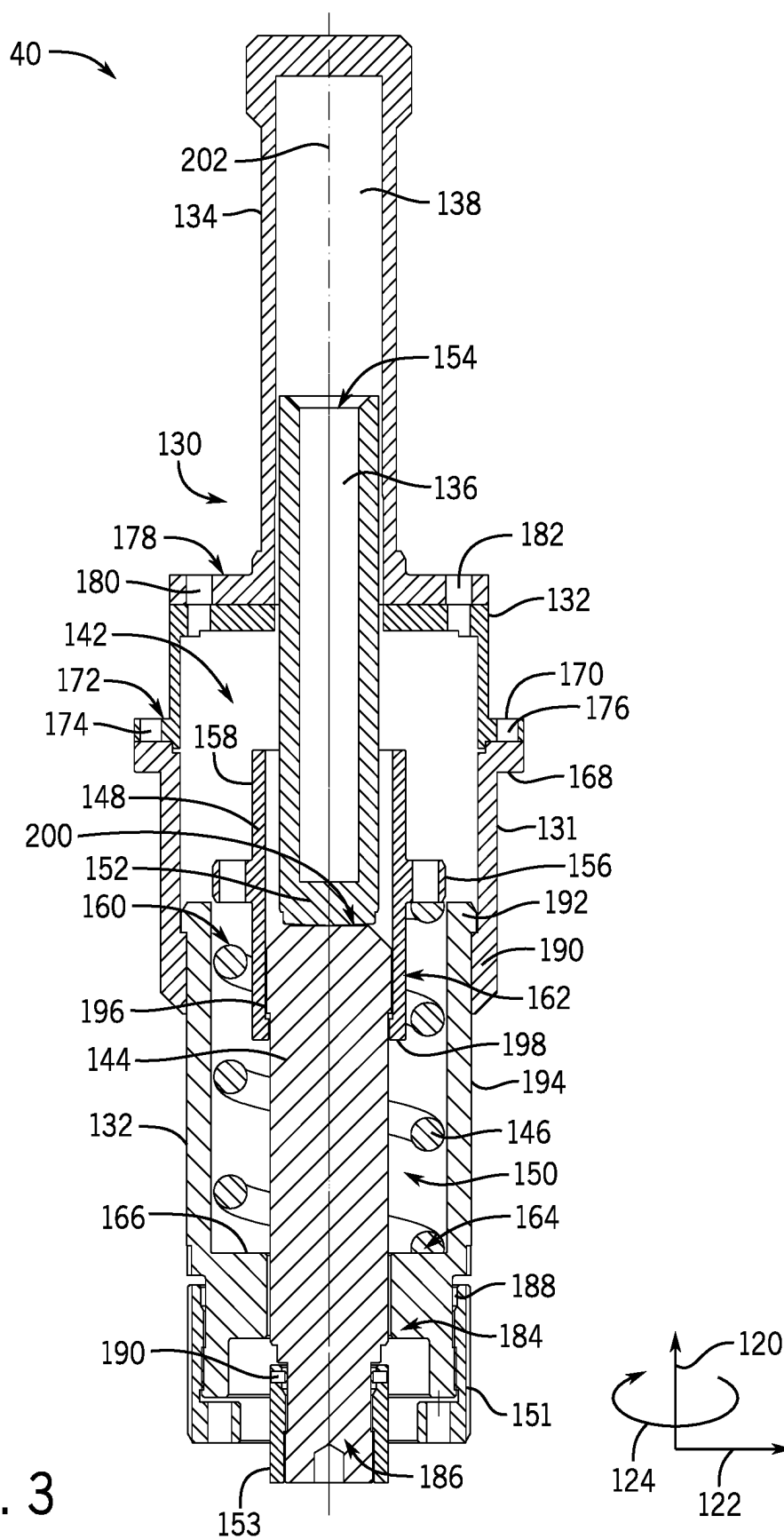


FIG. 3

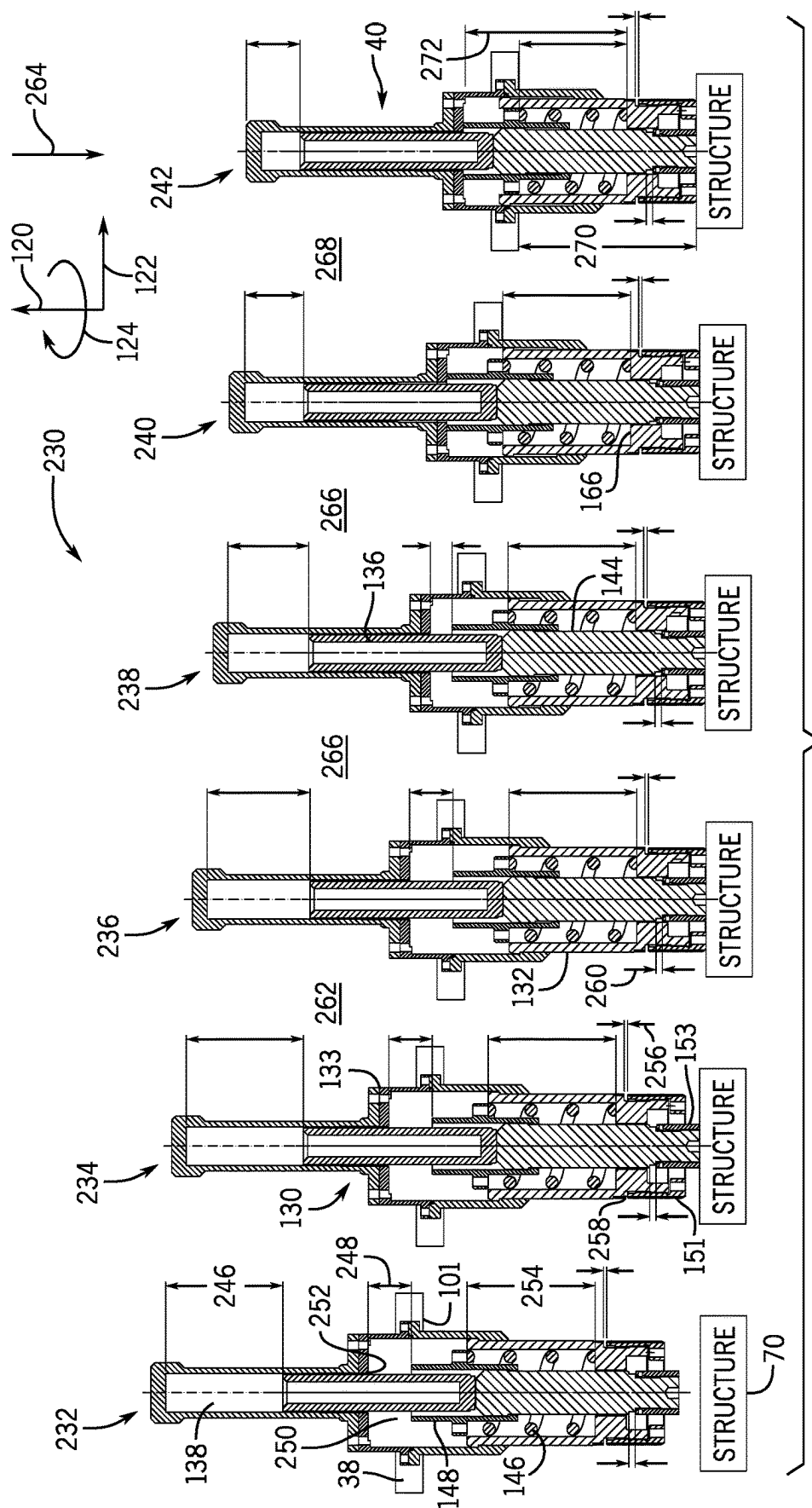


FIG. 4

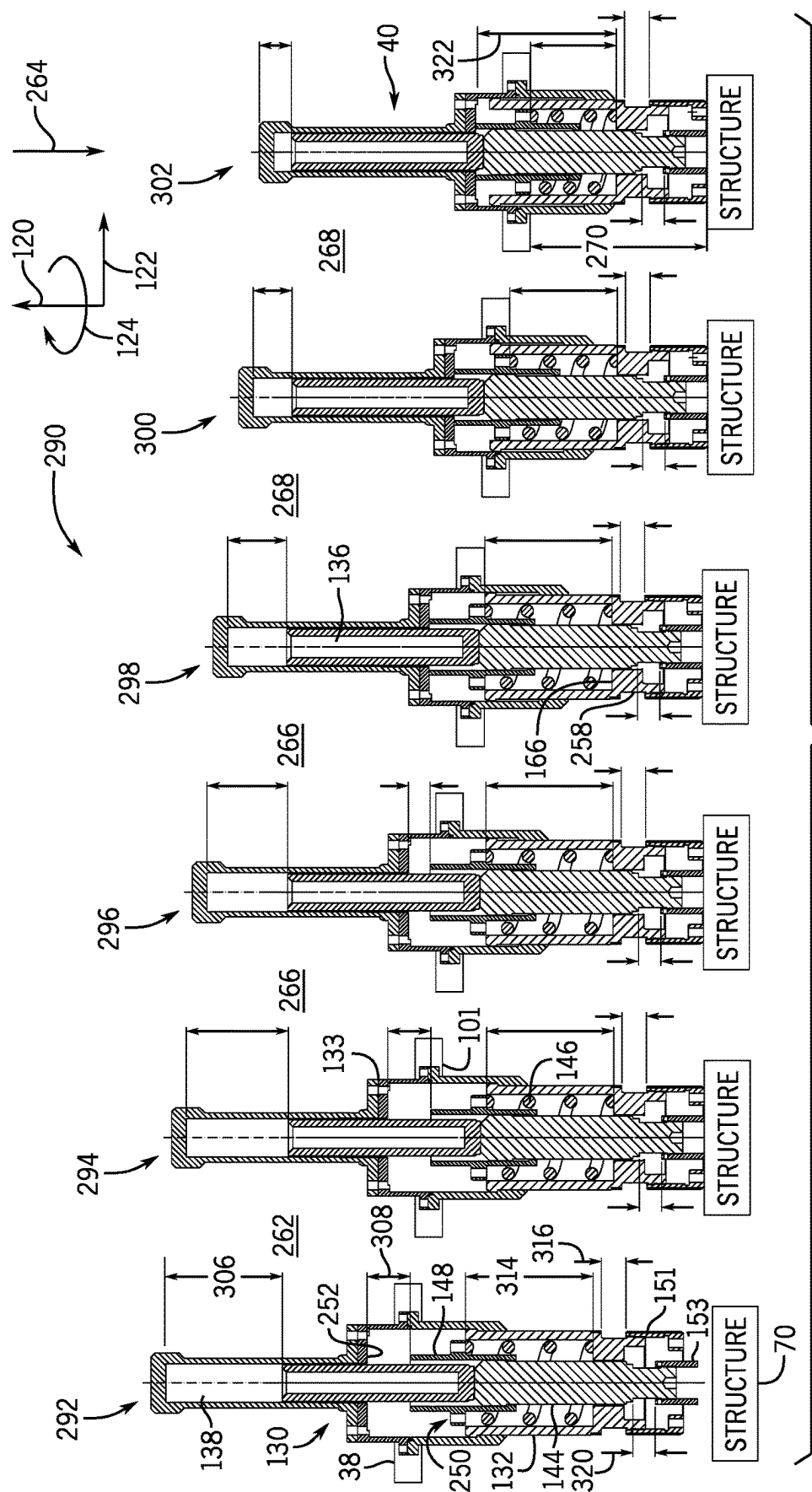


FIG. 5

SYSTEM AND METHOD FOR LANDING RETRIEVABLE PROCESS MODULE

BACKGROUND

[0001] The present disclosure generally relates to a landing system for a retrievable process module used in subsea systems.

[0002] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it may be understood that these statements are to be read in this light, and not as admissions of prior art.

[0003] In subsea applications, a retrievable module (e.g., retrievable process module, retrievable flow module, etc.) may be added to or retrieved from one or more subsea structures of a subsea system. The retrievable module includes a reconfigurable interior that may be altered based on the process to be performed by the retrievable module. However, due to the reconfiguration of internal components, a center of gravity of the retrievable module may be changed, which may inadvertently induce a bending moment on the retrievable module when placed on the subsea structure. Accordingly, a system for offsetting this induced bending moment is desired.

SUMMARY

[0004] A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

[0005] In certain embodiments, a system includes a landing system configured to land a retrievable module on a subsea structure. The landing system includes a landing post, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0006] In certain embodiments, a system includes a retrievable module and a landing system configured to land the retrievable module on a subsea structure. The landing system includes a landing post including one or more compensators configured to move in response to a landing force, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0007] In certain embodiments, a method including landing a retrievable module on a subsea structure via a landing system, including moving one or more compensators of a landing post in response to a landing force, and adjusting a position of the landing post relative to a landing side of the retrievable module via a landing adjuster.

[0008] Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to

one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended only to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0010] FIG. 1 is a schematic view of a subsea production system having at least one subsea structure that includes a retrievable module, according to an embodiment of the present disclosure;

[0011] FIG. 2 is a schematic view of one or more landing systems facilitating a soft landing of a retrievable module of FIG. 1 having varying mass distributions, according to an embodiment of the present disclosure;

[0012] FIG. 3 is a cross-sectional view of the landing system of FIG. 2, according to an embodiment of the present disclosure;

[0013] FIG. 4 is a schematic cross-sectional view of a landing sequence of the landing system of FIG. 2 having a low loading configuration, according to an embodiment of the present disclosure; and

[0014] FIG. 5 is a schematic cross-sectional view of a landing sequence of the landing system of FIG. 2 having a high loading configuration, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0015] Certain embodiments commensurate in scope with the present disclosure are summarized below. These embodiments are not intended to limit the scope of the disclosure, but rather these embodiments are intended only to provide a brief summary of certain disclosed embodiments. Indeed, the present disclosure may encompass a variety of forms that may be similar to or different from the embodiments set forth below.

[0016] As used herein, the term “coupled” or “coupled to” may indicate establishing either a direct or indirect connection (e.g., where the connection may not include or include intermediate or intervening components between those coupled), and is not limited to either unless expressly referenced as such. The term “set” may refer to one or more items. Wherever possible, like or identical reference numerals are used in the figures to identify common or the same elements. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale for purposes of clarification.

[0017] Furthermore, when introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment,” “an embodiment,” or “some embodiments” of the present disclosure are not intended to

be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, the phrase A “based on” B is intended to mean that A is at least partially based on B. Moreover, unless expressly stated otherwise, the term “or” is intended to be inclusive (e.g., logical OR) and not exclusive (e.g., logical XOR). In other words, the phrase A “or” B is intended to mean A, B, or both A and B.

[0018] The present disclosure is generally directed toward a landing system (e.g., soft landing system, spring system, etc.) for a retrievable module used in subsea applications (e.g., below or under the surface of a body of water, such as the sea, ocean, lake river, or the like). As discussed above, the landing system disclosed herein is adjustable to a retrievable module of variable mass density. The landing system is configured to couple to the retrievable module (e.g., retrievable module hub) and make initial contact with a subsea structure, base, or floor. During a landing of the retrievable module onto a subsea structure, the landing system is compressed in multiple (e.g., three) separate stages to enable retrievable modules of varying mass densities to land onto the subsea structure. The first two compression stages of the landing system (e.g., the soft landing) include contacting the subsea structure and compression of a first compressible portion (e.g., a damper) so as to maintain contact with the subsea structure. The third compression stage involves a mass of the retrievable module causing compression of a second compressible portion (e.g., a spring), such that the second compressible portion bears the mass of the retrievable module (e.g., localized mass).

[0019] As such, in certain embodiments of the present disclosure, the landing system includes a landing system that compresses in multiple stages. By the end of a first compression stage, both inner and outer extension nuts disposed on a bottom side of the landing system contact the subsea structure. During a second compression stage, a resistive pressure is applied by the first compressible portion to provide a soft-landing phase after initially making contact. During a third compression stage, the load of the retrievable module is placed on the second compressible portion.

[0020] In certain embodiments of the present disclosure, the landing system includes the inner and outer extension nuts disposed on the axial end of the landing system that contacts the subsea structure. The inner and outer extension nuts are axially coupled to the landing system and are configured to elongate the landing system to compensate for increased and/or decreased compression of the first and/or second compressible portions due to variance in the load distribution of the retrievable module.

[0021] With the foregoing in mind, FIG. 1 is a schematic view of a subsea system 10 having various subsea equipment that may receive the retrievable module using the landing system described in further detail below. As illustrated, the subsea system includes electrical cables 12 used for transmitting information and primary electrical power for various subsea components (e.g., actuators, sensors, etc.). The subsea system 10 may include a subsea hydrocarbon production system configured to extract oil or gas from a subterranean reservoir, a subsea fluid injection system configured to inject fluid (e.g., liquid or gas) into a subterranean reservoir, or any other subsea system associated with subterranean reservoirs. In certain embodiments, the subsea system 10 may include a subsea tree 14 coupled to a wellhead 16 to form a subsea station 18 configured to

extract and/or inject fluids relative to a subterranean reservoir. For example, the subsea station 18 may be configured to extract formation fluid, such as oil and/or natural gas, from the sea floor 20 through the well 22. In some embodiments, the subsea system 10 may include multiple subsea stations 18 that extract and/or inject fluids relative to respective wells 22.

[0022] In embodiments of the subsea system 10 configured for production, after passing through the subsea tree 14, the formation fluid flows through fluid conduits or pipes 24 to a pipeline manifold 26. The pipeline manifold 26 may connect to one or more flowlines 28 to enable the formation fluid to flow from the wells 22 to a surface platform 30. In some embodiments, the surface platform 30 may include a floating production, storage, and offloading unit (FPSO) or a shore-based facility. In addition to flowlines 28 that carry the formation fluid away from the wells 22, the subsea system 10 may include lines or conduits 32 that supply fluids, as well as carry control and data lines to the subsea equipment. These conduits 32 connect to a distribution module 34, which in turn couples to the subsea stations 18 via supply lines 36. In some scenarios, the platform 30 may be located a significant distance (e.g., greater than 100 m, greater than 1 km, greater than 10 km, or greater than 60 km) away from the wells 22. As discussed in further detail below, the subsea system 10 (e.g., the subsea tree 14, the subsea station 18, the pipeline manifold 26, and/or the distribution module 34) may include a connection or docking area, wherein a retrievable module 38 may be configured to land and connect with the subsea system 10. While the subsea system 10 described above is for extracting hydrocarbons, it should be understood that the present disclosure may also apply to other types of subsea systems 10 such as subsea injection systems (e.g., subsea gas injection system, subsea water injection system, and/or subsea carbon dioxide injection system).

[0023] In certain embodiments, at least one of the subsea stations 18 of the subsea system 10 may include the retrievable module 38 that may be retrievable from the subsea station 18. The retrievable module 38 may be configured to attach to the tree 14 or, in certain embodiments, other structures of the subsea system 10, such as the pipeline manifold 26 and/or the distribution module 34. As discussed in further detail herein, the retrievable module 38 may include one or more landing systems 40 to assist in orienting the retrievable module 38 relative to the subsea station 18 and making various connections, such as fluid connections, electrical power connections, communication and control connections, or any combination thereof. It may be recognized that the subsea system 10 disclosed herein is just an example of many possible arrangements. The retrievable module 38 and the landing systems 40 may be compatible with any number of arrangements of the subsea system 10. For example, the subsea system 10 may or may not include the platform 30.

[0024] FIG. 2 is a schematic view of one or more landing systems 40 facilitating a soft landing of a retrievable module 38 having varying mass distributions. In the illustrated embodiment, the retrievable module 38 includes components 60 having a center of gravity 62 disposed within an interior 64 of the retrievable module 38. As shown, the retrievable module 38 includes an arm portion 66 that is designed to engage a capture envelope 68 (e.g., a docking area or volume having various connections, such as

mechanical connections, electrical connections, fluid connections, etc.) of a subsea structure 70 (e.g., subsea station, subsea tree, subsea manifold etc.). The retrievable module 38 may include a reusable exterior 71 (e.g., housing, body, or framework) that may be outfitted in the interior 64 with components 60 of varying masses, thereby varying the mass distribution of the retrievable module 38. In certain embodiments, the components 60 may include one or more valves (e.g., gate valves, ball valves, check valves, chokes, blowout preventers, etc.), pumps, compressors, flowmeters, sensors (e.g., pressures sensors, temperature sensors, etc.), chemical injection systems (e.g., chemical injection metering valves), fluid intervention systems, subsea sampling systems, tree conversion systems (e.g., producer to injector), high-integrity pressure protection systems (HIPPS), fluid conduits, fluid tanks, accumulators, actuators, chokes, controllers (e.g., processor-based controllers having one or more processors, memory, and instructions to control the components), or any combination thereof.

[0025] As shown in view 72, a first retrievable module 74 includes first components 76 having a first center of gravity 78 at a first location 80. The first retrievable module 74 also includes a first arm portion 81. The first retrievable module 74 is oriented at a first orientation 82 (e.g., pose) relative to the subsea structure 70. In view 84, a second retrievable module 86 includes second components 88 having a second center of gravity 90 at a second location 92. The second retrievable module 86 also includes a second arm portion 93. As shown, the second retrievable module 86 has a second orientation 94 relative to the subsea structure 70. In the illustrated embodiment, the first orientation 82 and the second orientation 94 are different from one another due to the variance in mass distribution (e.g., differences between the first center of gravity 78 and the second center of gravity 90) of the first components 76 and the second components 88. As shown in the views 72 and 84, neither the first arm portion 81 of the first retrievable module 74 nor the second arm portion 93 of the second retrievable module 86 are positioned within the capture envelope 68 of the subsea structure 70. Although the illustrated embodiment shows the retrievable module 38 as having a certain shape, it may be recognized that the retrievable module 38 may have a variety of shapes.

[0026] As shown in view 96, the first retrievable module 74 is outfitted with a landing system 40 having landing posts 97 (e.g., landing posts 98, 100) on a bottom side 101 (e.g., landing side) of the retrievable module 74. Additionally, in view 102, the second retrievable module 86 is outfitted with the landing system 40 having the landing posts 97 (e.g., landing posts 104 and 106) on the bottom side 101 of the retrievable module 86. Both the landing posts 98, 100 and the landing posts 104, 106 respectively have the same positions with respect to the retrievable module 38. As shown, due to both the first retrievable module 74 and the second retrievable module 86 having the landing systems 40, both the first arm portion 81 of the first retrievable module 74 and the second arm portion 93 of the second retrievable module 86 are positioned within the capture envelope 68 of the subsea structure 70. Because both the first arm portion 81 and the second arm portion 93 are aligned within the capture envelope 68 despite the first retrievable module 74 and the second retrievable module 86 having centers of gravity 62 at different locations (e.g., different mass distributions), it may be appreciated that the landing

systems 40 may align (e.g., stabilize and orient an angle, a height, and a position of) retrievable modules 38 with different mass distributions with respect to the capture envelope 68. In this manner, the landing systems 40 enable a successful landing (e.g., capture) of the retrievable module 38 on the subsea structure 70 irrespective of the center of gravity 62 (e.g., mass distribution) of the components 60 (e.g., internal components) of the retrievable module 38, thereby saving time and cost of having to redesign the retrievable module 38.

[0027] FIG. 3 is a cross-sectional view of the landing system 40. The landing system 40 may be described with respect to an axial direction 120 (e.g., axial axis), a radial direction 122 (e.g., radial direction), and a circumferential direction 124 (e.g., circumferential axis). In the illustrated embodiment, the landing system 40 includes an outer sleeve 130 disposed about an inner sleeve 132. As shown, the outer sleeve 130 includes a bottom section 131, a middle section 133 and a top section 134. The outer sleeve 130 is configured to slide past (e.g., slide over, telescope past) the inner sleeve 132. In other words, the outer and inner sleeves 130 and 132 may be coaxial or concentric sleeves that axially expand and contract relative to one another in the axial direction 120. Additionally, the landing system 40 includes a stop pin 136 (e.g., hollow shaft portion or soft-landing piston) and a first compressible portion 138 (e.g., compensator, fluid damper) disposed in a first interior 142 (e.g., chamber) of the outer sleeve 130. The landing system 40 also includes an axial core portion 144 (e.g., central shaft portion), a second compressible portion 146 (e.g., compensator, spring), and a stopper 148 (e.g., annular stopper) disposed in a second interior 150 of the inner sleeve 130. The second compressible portion 146 may include one or more springs, such as one or more mechanical springs (e.g., helical springs). In some embodiments, the second compressible portion 146 may include one or more fluid springs (e.g., pneumatic springs or shock absorbers) and/or one or more magnetic springs (e.g., magnetic compensator). In some embodiments, the second compressible portion 146 may include one or more mechanical springs, one or more fluid springs, one or more magnetic springs, or a combination thereof. As shown, the axial core portion 144, the second compressible portion 146, and the stopper 148 may also be at least partially disposed in the first interior 142 of the outer sleeve 130. The landing system 40 also includes an outer extension nut 151 (e.g., threaded nut, landing adjuster) coupled to the inner sleeve 132, and an inner extension nut 153 (e.g., threaded nut, landing adjuster) coupled to the axial core portion 144.

[0028] In the illustrated embodiment, the axial core portion 144 is disposed axially adjacent to a first axial end 152 of the stop pin 136. Additionally, the first compressible portion 138 is disposed axially adjacent to a second axial end 154 of the stop pin 136. As shown, the stopper 148 is disposed circumferentially about both the axial core portion 144 and the stop pin 136 and is able to axially slide past both the axial core portion 144 and the stop pin 136. The stopper 148 includes a protrusion 156 (e.g., annular protrusion or flange) from an outer surface 158 of the stopper 148. The protrusion 156 is configured to abut and compress a free axial end 160 of the second compressible portion 146.

[0029] As shown, the second compressible portion 146 is disposed about the axial core portion 144 and also about a lower stopper portion 162 of the stopper 148. A fixed axial

end 164 of the second compressible portion 146 is coupled to an interior axial surface 166 of the inner sleeve 132. The second compressible portion 146 (e.g., spring) is configured to be compressed by the stopper 148 into the second interior 150 of the inner sleeve 132. Although the illustrated embodiment shows the second compressible portion 146 as being a spring, it may be recognized that the second compressible portion 146 may include other compressible devices (e.g., hydraulics, dampers, etc.).

[0030] The first compressible portion 138 is disposed with the first interior 142 of the outer sleeve 130 and is configured to be compressed by the stop pin 136 (e.g., soft-landing piston). In certain embodiments, the first compressible portion 138 is less stiff than the second compressible portion 146. In the illustrated embodiment, the first compressible portion 138 is illustrated as being a damper (e.g., water-filled damper), however it may be recognized that the first compressible portion 138 may include other compressible devices (e.g., hydraulics, springs, etc.). The first compressible portion 138 also may include one or more fluid ports or openings to facilitate a controlled discharge of fluid (e.g., water) during a soft landing when the stop pin 136 (e.g., soft-landing piston) moves upwardly in the first compressible portion 138, while intaking fluid (e.g., water) when not soft landing.

[0031] As shown, a bottom flange 168 of the bottom section 131 of the outer sleeve 130 is coupled to a middle flange 170 of the middle section 133 of the outer sleeve 130 via fasteners (e.g., threaded fasteners, screws, or bolts) passing through a plurality of holes 172 (e.g., bolt holes). In certain embodiments, the plurality of holes 172 (e.g., holes 174 and 176) may enable a coupling the bottom section 131 and/or the middle section 133 of the outer sleeve 130 to the retrievable module, thereby coupling the landing system 40 to the retrievable module. Additionally, the middle section 133 of the outer sleeve 130 is configured to couple to the top section 134 of the outer sleeve 130 via fasteners (e.g., threaded fasteners, screws, or bolts) passing through a plurality of holes 178 (e.g., holes 180, 182).

[0032] In the illustrated embodiment, the outer extension nut 151 is coupled to a sleeve axial end 184 of the inner sleeve 132, and the inner extension nut 153 coupled to a core axial end 186 of the axial core portion 144. The outer extension nut 151 is configured to axially slide relative to the inner sleeve 132, and the inner extension nut 153 is configured to axially slide relative to the axial core portion 144. In certain embodiments, the outer extension nut 151 may be threaded onto the sleeve axial end 184 of the inner sleeve 132, and the inner extension nut 153 may be threaded onto the core axial end 186 of the axial core portion 144. An outer lock nut 188 may be disposed between the outer extension nut 151 and the inner sleeve 132 to lock a position of the outer extension nut 151 relative to the inner sleeve 132. Additionally, an inner lock nut 190 may be disposed between the inner extension nut 153 to lock a position of the inner extension nut 153 relative to the axial core portion 144. In certain embodiments, a specialized tool (e.g., specialized wrench) may be used to adjust the outer extension nut 151 and/or the inner extension nut 153. The outer extension nut 151 and/or the inner extension are configured to contact the subsea structure during the landing of the retrievable module 38.

[0033] In the illustrated embodiment, the bottom portion 131 of the outer sleeve 130 includes an inward protrusion

190 (e.g., inner annular protrusion) configured to interlock with an outward protrusion 192 (e.g., outer annular protrusion) disposed on an outer surface 194 of the inner sleeve 132. As shown, the outward protrusion 192 and the inward protrusion 190 are configured to axially block the inner sleeve 132 from continuing to slide axially relative to the outer sleeve 130, so that the inner sleeve 132 remains axially secured or captured within the outer sleeve 130.

[0034] In the illustrated embodiment, the axial core portion 144 includes an outward core protrusion 196 (e.g., outer annular protrusion) and the stopper 140 includes an inward stopper protrusion 198 (e.g., inner annular protrusion). As shown, the outward core protrusion 196 and the inward stopper protrusion 198 are configured to axially block the stopper 140 from axially sliding off the axial core portion 144. Additionally, the protrusion 156 of the stopper 140 axially abuts the second compressible portion 146, ensuring that the stopper 140 remains disposed over the interface 200 (e.g., axial abutment) between the axial core portion 144 and the stop pin 136.

[0035] In the illustrated embodiment, the outer sleeve 130, the inner sleeve 132, the stop pin 136, the first compressible portion 138, the stopper 140, the axial core portion 144, the second compressible portion 146, the outer extension nut 151, and the inner extension nut 153 are shown as being annular in shape and disposed about a central axis 202. In certain embodiments, the outer sleeve 130, the inner sleeve 132, the stop pin 136, the first compressible portion 138, the stopper 140, the axial core portion 144, the second compressible portion 146, the outer extension nut 151, the inner extension nut 153, or a combination thereof, may not be annular in shape and/or may be radially offset from the central axis 202.

[0036] FIG. 4 is a schematic cross-sectional view of a landing sequence 230 of the landing system 40 while supporting a retrievable module 38 (e.g., portion of a retrievable module 38) having a low mass (e.g., low mass density). In the illustrated embodiment, the landing system 40 is directly coupled to the retrievable module 38 (e.g., retrievable module hub) and is being lowered onto the subsea structure 70. As discussed herein, the subsea structure 70 may include a subsea tree, or any other subsea structure used in a subsea system. As shown, the landing sequence 230 includes views 232, 234, 236, 238, 240, and 242, which represent the landing system 40 chronologically during a landing of the retrievable module 38 onto the subsea structure 70. It may be recognized that these views may occur in the reverse order during a retrieval of the retrievable module 38 from the subsea structure 70. Each view includes three dimensions, including a first length 246 of the first compressible portion 138, a distance 248 spanning from a free axial end 250 of the stopper 148 and an interior axial surface 252 of the middle portion 132 of the outer sleeve 130, and a second length 254 of the second compressible portion 146.

[0037] Additionally, each of the views includes an overhang distance 256 between an axial end 258 of the outer extension nut 151 and the inner sleeve 132. In certain embodiments, the overhang distance 256 between the axial end 258 of the outer extension nut 151 and the inner sleeve 132 is equivalent to an overhang distance 260 between the inner extension nut 153 and the axial core portion 144, since the outer extension nut 151 and the inner extension nut 153 may be adjusted by the same axial distance. Because the outer extension nut 151 does not slide relative to the inner

sleeve 132 during the landing, the overhang distance 256 remains the same during the landing sequence 230. The same holds true for the overhang distance 260.

[0038] In the view 232, the landing system 40 coupled to the retrievable module 38 is approaching the subsea structure 70 in a vertical downward direction. As discussed in more detail herein, because the retrievable module 38 has a minimal (e.g., low) mass density in the landing sequence 230, the overhang distance 256 (e.g., and overhang distance 260) are set to a low value. In the view 232, neither the first compressible portion 138 nor the second compressible portion 146 are compressed.

[0039] In the view 234, the inner extension nut 153 contacts the subsea structure 70. The first length 246 of the first compressible portion 138, the distance 248 spanning from a free axial end 250 of the stopper 148 and the interior axial surface 252 of the bottom portion 131 of the outer sleeve 130, and the second length 254 of the second compressible portion 146 have not changed from the view 232. The inner extension nut 153 contacting the subsea structure 70 marks the beginning of a first compression stage 262 of the landing system 240. During the first compression stage 262, the axial core portion 144 drives the stop pin 136 (e.g., soft-landing piston) to compress and/or discharge fluid (e.g., water) in the first compressible portion 138. Additionally, the inner sleeve 132 simultaneously slides axially past the axial core portion 144 in the direction 264, wherein the direction 264 is opposite of the axial direction 120.

[0040] In the view 236, the both the inner extension nut 153 and the outer extension nut 151 are in contact with the subsea structure 70, marking the end of the first compression stage 262 and the beginning of the second compression stage 266. As shown, by the end of the first compression stage 262, the first length 246 of the first compressible portion 138 has decreased due to compression of the first compressible portion 138. However, the distance 248 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 of the bottom portion 131 of the outer sleeve 130, and the second length 254 of the second compressible portion 146 have not changed from the view 234.

[0041] In the view 238, the landing system 40 is shown as undergoing the second compression stage 266. During the second compression stage 266, the axial core portion 144 continues to drive the stop pin 136 to compress and/or discharge fluid (e.g., water) in the first compressible portion 138. Additionally, during the second compression stage 266, the outer sleeve 130 is configured simultaneously to slide toward (but not yet contact) the stopper 148 due to the outer sleeve 130 sliding over the inner sleeve 132. As shown, during the second compression stage 266, the first length 246 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 248 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 has decreased due to outer sleeve 130 sliding over the inner sleeve 132. However, the second length 254 of the second compressible portion 146 has not changed from the view 236. The second compressible stage 266 may also be known as a soft-landing stage of the landing system 40.

[0042] In the view 240, the outer sleeve 130 contacts the stopper 148, thereby marking an end of the second compression stage 266 and a beginning of the third compression stage 268. During the third compression stage 268, the axial

core portion 144 continues to compress the first compressible portion 138. Additionally, during the third compression stage 268, the outer sleeve 130 simultaneously drives the stopper 148 to compress the second compressible portion 146 (e.g., spring) via the protrusion 156 of the stopper 148. As shown, by the beginning of the third compression stage 268, the first length 246 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 248 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 is now zero due to contact between the interior axial surface 252 and the stopper 148. However, the second length 254 of the second compressible portion 146 (e.g., spring) has not changed from the view 238.

[0043] In the view 242, the outer sleeve 130, which is coupled to the retrievable module 38, continues to drive the stopper 148 to compress the second compressible portion 146 (e.g., spring). As shown, the outer sleeve 130 has also continued to slide over (e.g., past, relative to) the inner sleeve 132. The axial core portion 144 has simultaneously continued to compress the first compressible portion 138. As shown, by the end of the third compression stage 268, the first length 246 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 248 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 remains zero due to contact between the interior axial surface 252 and the stopper 148. Furthermore, the second length 254 of the second compressible portion 146 (e.g., spring) has decreased due to compression of the second compressible portion 146.

[0044] At the end of the third compression stage 268, the retrievable module 38 is separated by a distance 270 from the subsea structure 70. Additionally, at the end of the third compression stage 268 (as shown in the view 242), the interior axial surface 250 of the outer sleeve 130 is separated from the interior axial surface 166 of the inner sleeve 132 by a distance 272. As shown, between the view 232 and the view 242, the inner sleeve 132, the stop pin 136, the first compressible portion 138, the axial core portion 144, the second compressible portion 146, and the stopper 148 have moved relative to the outer sleeve 130, which is fixed to the retrievable module 38.

[0045] FIG. 5 is a schematic cross-sectional view of a landing sequence 290 of the landing system 40 while supporting a retrievable module 38 (e.g., portion of a retrievable module 38) having a higher mass (e.g., higher mass density) compared to the landing sequence shown in FIG. 4. In the illustrated embodiment, the landing system 40 is directly coupled to the retrievable module 38 (e.g., via fasteners) and is being lowered onto the subsea structure 70. As discussed herein, the subsea structure 70 may include a subsea tree, or any other subsea structure used in a subsea system. As shown, the landing sequence 290 includes views 292, 294, 296, 298, 300, and 302, which represent the landing system 40 chronologically during a landing of the retrievable module 38 onto the subsea structure 70. It may be recognized that these views may occur in the reverse order during a retrieval of the retrievable module 38 from the subsea structure 70. Each view includes three dimensions, including a first length 306 of the first compressible portion 138, a distance 308 spanning from a free axial end 250 of the stopper 148 and an interior axial surface 252 of the middle

portion 132 of the outer sleeve 130, and a second length 314 of the second compressible portion 146.

[0046] Additionally, the view 292 includes an overhang distance 316 between the axial end 258 of the outer extension nut 151 and the inner sleeve 132. In certain embodiments, the overhang distance 316 between the axial end 258 of the outer extension nut 151 and the inner sleeve 132 is equivalent to an overhang distance 320 between the inner extension nut 153 and the axial core portion 144, since the outer extension nut 151 and the inner extension nut 153 may be adjusted by the same axial distance. Because the outer extension nut 151 does not slide relative to the inner sleeve 132 during the landing, the overhang distance 316 remains the same during the landing sequence 290. The same holds true for the overhang distance 320. As shown, the overhang distances 316 and 320 in the landing sequence 290 are greater than the respective overhang distances shown in the landing sequence 230. As discussed further herein, the increase in the overhang distances 316 and 320 is to compensate for the increased mass of the retrievable module 38.

[0047] In the view 292, due to the increased overhang distances 316 and 320, the inner extension nut 153 is already in contact with the subsea structure 70. The inner extension nut 153 contacting the subsea structure 70 marks the beginning of the first compression stage 262 of the landing system 40. During the first compression stage 262, the axial core portion 144 drives the stop pin 136 (e.g., soft-landing piston) to compress and/or discharge fluid (e.g., water) in the first compressible portion 138. Additionally, the inner sleeve 132 simultaneously slides axially past the axial core portion 144 in the direction 264, wherein the direction 264 is opposite of the axial direction 120.

[0048] In the view 294, both the inner extension nut 153 and the outer extension nut 151 are in contact with the subsea structure 70, marking the end of the first compression stage 262 and the beginning of the second compression stage 266. As shown, by the end of the first compression stage 262, the first length 306 of the first compressible portion 138 has decreased due to compression of the first compressible portion 138. However, the distance 308 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 of the bottom portion 131 of the outer sleeve 130, and the second length 314 of the second compressible portion 146 have not changed from the view 292.

[0049] In the view 296, the landing system 40 is shown as undergoing the second compression stage 266. During the second compression stage 266, the axial core portion 144 continues to drive the stop pin 136 to compress and/or discharge fluid (e.g., water) in the first compressible portion 138. Additionally, during the second compression stage 266, the outer sleeve 130 is configured to simultaneously slide toward (but not yet contact) the stopper 148 due to the outer sleeve 130 sliding over the inner sleeve 132. As shown, during the second compression stage 266, the first length 306 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 308 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 has decreased due to outer sleeve 130 sliding over the inner sleeve 132. However, the second length 314 of the second compressible portion 146 has not changed from the view 294. The second compressible stage 266 may also be known as a soft-landing stage of the landing system 40.

[0050] In the view 298, the outer sleeve 130 contacts the stopper 148, thereby marking an end of the second compression stage 266 and a beginning of the third compression stage 268. During the third compression stage 268, the axial core portion 144 continues to compress the first compressible portion 138. Additionally, during the third compression stage 268, the outer sleeve 130 simultaneously drives the stopper 148 to compress the second compressible portion 146 (e.g., spring) via the protrusion 156 of the stopper 148. As shown, by the beginning of the third compression stage 268, the first length 306 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 308 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 is now zero due to contact between the interior axial surface 252 and the stopper 148. However, the second length 314 of the second compressible portion 146 (e.g., spring) has not changed from the view 296.

[0051] In the view 300, the landing system 40 is shown as undergoing the third compression stage 268. As shown, the outer sleeve 130, which is coupled to the retrievable module 38, continues to drive the stopper 148 to compress the second compressible portion 146 (e.g., spring). As shown, the outer sleeve 130 has also continued to slide over (e.g., past, relative to) the inner sleeve 132. The axial core portion 144 has simultaneously continued to compress the first compressible portion 138. As shown, the first length 306 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 308 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 remains zero due to contact between the interior axial surface 252 and the stopper 148. Furthermore, the second length 314 of the second compressible portion 146 (e.g., spring) has decreased due to compression of the second compressible portion 146.

[0052] In the view 302, the outer sleeve 130 continues to drive the stopper 148 to compress the second compressible portion 146. As shown, the outer sleeve 130 has also continued to slide over (e.g., past, relative to) the inner sleeve 132. The axial core portion 144 has simultaneously continued to compress the first compressible portion 138. As shown, by the end of the third compression stage 268, the first length 306 of the first compressible portion 138 has continued to decrease due to compression of the first compressible portion 138. Additionally, the distance 308 spanning from the free axial end 250 of the stopper 148 and the interior axial surface 252 remains zero due to contact between the interior axial surface 252 and the stopper 148. Furthermore, the second length 314 of the second compressible portion 146 has decreased due to compression of the second compressible portion 146 (e.g., spring).

[0053] In the illustrated embodiment, in the view 302 (e.g., at the end of the third compression stage 268), the first length 306 of the first compressible portion 138 is smaller than the first length 246 of the first compressible portion 138 shown in FIG. 4. Additionally, as shown, the second length 314 of the second compressible portion 146 is smaller than the second length 254 of the second compressible portion 146 shown in FIG. 4. That is, after completion of the third compression 268, the lengths of the first compressible portion 138, the second compressible portion 146, or a combination thereof, are based on a mass of the load (e.g., the

retrievable module 38) supported by the landing system 40. For example, in response to the retrievable module 38 having a higher mass density, the lengths of the first compressible portion 138, the second compressible portion 146, or the combination thereof, is configured to decrease. Additionally, at the end of the third compression stage 268 (as shown in the view 302), the interior axial surface 250 of the outer sleeve 130 is separated from the interior axial surface 166 of the inner sleeve 132 by a distance 322. As shown, compared to the distance 272 of FIG. 4, the distance 322 between the inner sleeve 132 and the outer sleeve 130 has decreased. As shown in the view 302, the retrievable module 38 is separated by the distance 270 from the subsea structure 70 after completion of the third compression stage 268.

[0054] As shown, both the overhang distance 316 of the outer extension nut 151 and the overhang distance 320 of the inner extension nut 153 are greater than the overhang distances 256 and 260 (shown in FIG. 4), respectively. In the illustrated embodiment, the amount of increase in overhang of the outer extension nut 151 and the inner extension nut 153 matches the amount of decrease in the distance 322 between the interior axial surfaces 250 and 166. That is, the overhang distances 316 and 320 compensate for the decrease in the distance 322 (e.g., increase in overlap between the inner sleeve 132 and the outer sleeve 130) due to the increase load of the retrievable module 38. It may be appreciated that the outer extension nut 151 and/or the inner extension nut 153 are configured to adjust the position of the landing post 40 relative to the landing side 101 of the retrievable module 38 in a first crosswise direction (e.g., vertical direction 264) to the landing side 101. In certain embodiments, the outer extension nut 151 and/or the inner extension nut 153 may be configured to adjust the position of the landing post 40 relative to the landing side 101 of the retrievable module 38 in a second crosswise direction (e.g., direction 122) to the landing side 101. That is, the outer extension nut 151 (e.g., landing adjuster) and/or the inner extension nut 153 (e.g., landing adjuster) may compensate (e.g., via the overhang of the extension nuts 151 and 153) for the change in overlap (e.g., position) of the inner sleeve 132 and the outer sleeve 130 to stabilize landing of retrievable modules 38 of different masses, centers of gravity, or a combination thereof.

[0055] Technical effects of the disclosed embodiments include a landing system designed to attach to a retrievable module and offset a moment induced onto a retrievable module due to an imbalance in mass. The landing system includes a soft-landing compressible portion for making initial contact and a load-bearing compressible portion that bears the load of the retrievable module. The load-bearing compressible portion and/or the soft-landing compressible portion may be displaced by varying amounts based on the mass of the retrievable module. To offset the displacement of either the load-bearing compressible portion or the soft-landing compressible portion, the landing system includes inner and outer extension nuts disposed on an axial end of the landing system that may be adjusted to compensate for the different compression distances. As such, the landing system may be adjusted based on the mass distribution of the retrievable module to ensure that retrievable modules of variable mass distribution may land on a subsea structure while avoiding an induced bending moment due to a mass imbalance, thereby avoiding the expenditure of time and

money of having to redesign the retrievable module each time the mass distribution of the retrievable module is reconfigured.

[0056] The subject matter described in detail above may be defined by one or more clauses, as set forth below.

[0057] A system includes a landing system configured to land a retrievable module on a subsea structure. The landing system includes a landing post, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0058] The system of the preceding clause, wherein the landing post includes one or more compensators configured to move in response to a landing force.

[0059] The system of any preceding clause, wherein the one or more compensators include one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof.

[0060] The system of any preceding clause, wherein the one or more compensators include a plurality of compensators configured to actuate in a plurality of stages.

[0061] The system of any preceding clause, wherein the plurality of compensators includes a fluid compensator including a piston disposed in a chamber and a mechanical compensator including a spring.

[0062] The system of any preceding clause, including a plurality of landing systems configured to land the retrievable module on the subsea structure, wherein each of the plurality of landing systems includes the landing post and the landing adjuster.

[0063] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

[0064] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side.

[0065] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a first direction crosswise to the landing side.

[0066] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a second direction along the landing side.

[0067] The system of any preceding clause, wherein the first direction is vertical, and the second direction is horizontal.

[0068] The system of any preceding clause, wherein the landing adjuster includes a manual landing adjuster, an automatic landing adjuster, or a combination thereof.

[0069] The system of any preceding clause, wherein the landing adjuster includes one or more threaded landing adjusters.

[0070] The system of any preceding clause, including the retrievable module including one or more of a valve, a pump, a compressor, a flowmeter, a sensor, a chemical injection system, a fluid intervention system, a subsea sampling system, a tree conversion system, a high-integrity pressure protection system, a fluid conduit, a fluid tank, an accumulator, an actuator, a choke, a controller, or any combination thereof.

[0071] The system of any preceding clause, including the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

[0072] A system including a retrievable module and a landing system configured to land the retrievable module on a subsea structure. The landing system includes a landing post including one or more compensators configured to move in response to a landing force, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0073] The system of the preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

[0074] The system of any preceding clause, wherein the one or more compensators include one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof, wherein the one or more compensators are configured to actuate in one or more stages.

[0075] The system of any preceding clause, including the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

[0076] A method including landing a retrievable module on a subsea structure via a landing system, including moving one or more compensators of a landing post in response to a landing force, and adjusting a position of the landing post relative to a landing side of the retrievable module via a landing adjuster.

[0077] As used herein, the terms “inner” and “outer”; “up” and “down”; “upper” and “lower”; “upward” and “downward”; “above” and “below”; “inward” and “outward”; and other like terms as used herein refer to relative positions to one another and are not intended to denote a particular direction or spatial orientation. The terms “couple,” “coupled,” “connect,” “connection,” “connected,” “in connection with,” and “connecting” refer to “in direct connection with” or “in connection with via one or more intermediate elements or members.”

[0078] The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. Moreover, the order in which the elements of the methods described herein are illustrated and described may be re-arranged, and/or two or more elements may occur simultaneously. The embodiments were chosen and described to best explain the principals of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated.

[0079] Finally, the techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . .” or “step for [perform]ing [a function] . . .,”

it is intended that such elements are to be interpreted under 35 U.S.C. § 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. § 112(f).

1. A system, comprising:

a landing system configured to land a retrievable module on a subsea structure, the landing system comprising: an outer sleeve disposed about an inner sleeve; a landing post; and a landing adjuster coupled to an axial end of the inner sleeve and configured to adjust a position of the landing post relative to a landing side of the retrievable module.

2. The system of claim 1, wherein the landing post comprises one or more compensators configured to move in response to a landing force.

3. The system of claim 2, wherein the one or more compensators comprise one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof.

4. The system of claim 2, wherein the one or more compensators comprise a plurality of compensators configured to actuate in a plurality of stages.

5. The system of claim 4, wherein the plurality of compensators comprises a fluid compensator comprising a piston disposed in a chamber and a mechanical compensator comprising a spring.

6. The system of claim 2, comprising a plurality of landing systems configured to land the retrievable module on the subsea structure, wherein each of the plurality of landing systems comprises the landing post and the landing adjuster.

7. The system of claim 1, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

8. The system of claim 1, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side.

9. The system of claim 1, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a first direction crosswise to the landing side.

10. The system of claim 9, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a second direction along the landing side.

11. The system of claim 10, wherein the first direction is vertical, and the second direction is horizontal.

12. The system of claim 1, wherein the landing adjuster comprises a manual landing adjuster, an automatic landing adjuster, or a combination thereof.

13. The system of claim 1, wherein the landing adjuster comprises one or more threaded landing adjusters.

14. The system of claim 1, comprising the retrievable module including one or more of a valve, a pump, a compressor, a flowmeter, a sensor, a chemical injection system, a fluid intervention system, a subsea sampling system, a tree conversion system, a high-integrity pressure protection system, a fluid conduit, a fluid tank, an accumulator, an actuator, a choke, a controller, or any combination thereof.

15. The system of claim **14**, comprising the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

16. A system, comprising:

a retrievable module; and

a landing system configured to land the retrievable module on a subsea structure, the landing system comprising:

an outer sleeve disposed about an inner sleeve;

a landing post comprising one or more compensators configured to move in response to a landing force; and

a landing adjuster coupled to an axial end of the inner sleeve and configured to adjust a position of the landing post relative to a landing side of the retrievable module.

17. The system of claim **16**, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side, wherein the landing adjuster is configured to adjust the position to

stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

18. The system of claim **16**, wherein the one or more compensators comprise one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof, wherein the one or more compensators are configured to actuate in one or more stages.

19. The system of claim **16**, comprising the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

20. A method, comprising:

landing a retrievable module on a subsea structure via a landing system, comprising:

moving one or more compensators of a landing post in response to a landing force;

adjusting a position of the landing post in a first direction relative to a landing side of the retrievable module via a landing adjuster; and

adjusting the position of the landing post in a second direction relative to the landing side of the retrievable module via the landing adjuster.

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